

The Helix–Hilbert–Polya Bridge: Conditional Projection Closure

Samuel Lavery

Draft proof bridge, 2026-06-13

Abstract

This note records the mathematical core currently formalized in Lean and evaluated by the numerical model. The Hilbert–Polya side is stated as an unconditional infrastructure chain. The remaining conditional input is projection faithfulness: dimensional projection from the three-dimensional FTA-generated source fiber to the lower-dimensional readout must preserve fiber capture events and must not create events absent from the source fiber. The source side is geometric: $\pi/3$ placement, prime/residue channel counting, and harmonic phasor accumulation. The fiber is derived from that source. Certain parts of the derived fiber identify with conventional L , ζ , Hurwitz, and Euler–Maclaurin forms after projection; those identifications are coincidental representations, not source definitions. Under that hypothesis the formal closure gives GRH, and the zeta/L1 specialization gives RH through the existing zeta bridge.

Contents

1	Objects and Notation	2
2	The $\pi/3$ Helix Geometry	2
2.1	Canonical Scale	2
2.2	Non-principal Character Channels	3
2.3	The Principal/Zeta Channel	3
3	Standing Readout and Projection to One Dimension	4
3.1	Phase Conventions	4
4	Harmonic Cost and the Quantum Ladder	5
5	Hilbert–Polya Infrastructure: R1–R9	5
6	Projection Faithfulness	7
7	Conditional Closure	8
8	Mathematical Reproduction of the Computation	9
8.1	Placed Geometric Fiber	9
8.2	Residue Coordinates for the Geometric Tail	10
8.3	Principal One-Dimensional Tail Error Control	11
9	Formal and Computational Support	11
9.1	Formalization Map	12
9.2	Lean Verification Protocol	12

1 Objects and Notation

Let χ be a Dirichlet character modulo q and write

$$L(s, \chi)$$

for Mathlib's Dirichlet L -function. For non-principal χ , define the strip zero set

$$\text{NTZ}(\chi) := \{\rho \in \mathbb{C} : 0 < \text{Re } \rho < 1, L(\rho, \chi) = 0\}.$$

The per-character GRH predicate formalized in Lean is

$$\text{GRH}(\chi) := \forall \rho \in \text{NTZ}(\chi), \quad \text{Re } \rho = \frac{1}{2}.$$

The complete all-character predicate is

$$\text{GRH}_{\text{complete}} := \forall q, \forall \chi \pmod{q}, \quad \text{GRH}(\chi).$$

For zeta, let

$$\text{NTZ}_\zeta := \{\rho \in \mathbb{C} : 0 < \text{Re } \rho < 1, \zeta(\rho) = 0\}.$$

2 The $\pi/3$ Helix Geometry

2.1 Canonical Scale

The numerical model and the formal gauge use the canonical $\pi/3$ chart

$$A = \frac{\pi}{6}, \quad ds = \frac{\pi}{3}, \quad C = 2A \, ds = \left(\frac{\pi}{3}\right)^2.$$

The exact area-law placement is

$$R_n^2 = Cn, \quad R_n = \frac{\pi}{3}\sqrt{n}.$$

At the source level the data are the $\pi/3$ coordinates, the channel weights coming from residue counting, and the harmonic phase on the helix. The fiber is derived from that source. No one-dimensional continuation formula is used to create it. The standard symbols L , ζ , and Hurwitz tails below occur because parts of the derived fiber identify with those representations after the geometric fiber has already been placed. Thus the three-dimensional source fiber atom at height t , for $s = \frac{1}{2} + it$, is

$$a_n(s) = w(n) \frac{1}{R_n} \exp\{-it \text{Log}(R_n^2)\} = w(n)(Cn)^{-s}.$$

Lemma 2.1 (Area-law closed form). *For any channel weights $w(n)$ for which the corresponding Dirichlet series is defined,*

$$F_w(s) := \sum_{n \geq 1} w(n)(Cn)^{-s} = C^{-s} \sum_{n \geq 1} w(n)n^{-s}.$$

Proof. Use $R_n^2 = Cn$ and $s = \frac{1}{2} + it$:

$$R_n^{-1} e^{-it \text{Log}(R_n^2)} = (R_n^2)^{-1/2} (R_n^2)^{-it} = (R_n^2)^{-s} = (Cn)^{-s}.$$

Summing gives the result. □

2.2 Non-principal Character Channels

For a non-principal Dirichlet character χ modulo q , the channel weights are

$$w(n) = \chi(n).$$

Therefore the source channel sum is

$$F_\chi(s) = \sum_{n \geq 1} \chi(n) (Cn)^{-s}.$$

In the projected one-dimensional notation this same harmonic sum is

$$F_\chi(s) = C^{-s} L(s, \chi).$$

Residue-wise, with zero residues omitted, the same placed channel may be reindexed as

$$F_\chi(s) = C^{-s} q^{-s} \sum_{r \bmod q} \chi(r) \zeta\left(s, \frac{r}{q}\right).$$

For χ_3 ,

$$\chi_3(1) = 1, \quad \chi_3(2) = -1, \quad \chi_3(0) = 0,$$

and hence

$$F_{\chi_3}(s) = C^{-s} 3^{-s} \left[\zeta\left(s, \frac{1}{3}\right) - \zeta\left(s, \frac{2}{3}\right) \right].$$

2.3 The Principal/Zeta Channel

For the zeta/L1 channel the fiber is eta-regularized:

$$w(n) = (-1)^{n+1}.$$

The source channel sum is

$$F_{L1}(s) = \sum_{n \geq 1} (-1)^{n+1} (Cn)^{-s}.$$

In the projected one-dimensional notation the eta factor is

$$\eta(s) = \sum_{n \geq 1} (-1)^{n+1} n^{-s} = (1 - 2^{1-s}) \zeta(s).$$

Thus the L1 fiber is

$$F_{L1}(s) = C^{-s} \eta(s) = C^{-s} (1 - 2^{1-s}) \zeta(s).$$

Equivalently, in residue form modulo 2,

$$F_{L1}(s) = C^{-s} 2^{-s} \left[\zeta\left(s, \frac{1}{2}\right) - \zeta(s, 1) \right].$$

On the critical line $\text{Re } s = \frac{1}{2}$, the eta factor is nonzero:

$$1 - 2^{1-s} \neq 0,$$

because $2^{1-s} = 1$ would imply $2^{1-\text{Re } s} = 1$, hence $\text{Re } s = 1$.

3 Standing Readout and Projection to One Dimension

This is the layer where the placed three-dimensional fiber is read in the lower-dimensional coordinate chart. The functions below are readouts of the source geometry and channel counts, with the coordinate phase correction kept explicit.

For $s = \frac{1}{2} + it$, the projected complex fiber is

$$F_\chi(t) = F_\chi\left(\frac{1}{2} + it\right).$$

The real standing readout has the form

$$Z_\chi(t) = \text{Re} \left(e^{i\Theta_\chi(t)} F_\chi\left(\frac{1}{2} + it\right) \right),$$

where

$$\Theta_\chi(t) = \arg \Gamma\left(\frac{1/2 + a}{2} + \frac{it}{2}\right) + \frac{t}{2} \log \frac{q}{\pi} + t \log C + \alpha_\chi.$$

Here $a \in \{0, 1\}$ is the parity and α_χ is the character/root-number gauge. For the L1/zeta eta-regularized channel one subtracts the eta gauge

$$\arg(1 - 2^{1-s}).$$

The term $t \log C$ is not a new oscillation. It is the coordinate correction from

$$\text{Log}(R_n^2) = \text{Log}(Cn) = \log n + \log C.$$

Omitting it rotates the real readout into the wrong coordinate chart.

3.1 Phase Conventions

All logarithms of positive real quantities are real logarithms. Thus

$$\text{Log}(R_n^2) = \log(Cn), \quad \log q, \quad \log C$$

use the real branch. The gamma phase in the standing readout is the continuous phase

$$\vartheta_\chi(t) = \text{Im} \log \Gamma\left(\frac{1/2 + a + it}{2}\right) + \frac{t}{2} \log \frac{q}{\pi},$$

with the branch chosen continuously along the vertical line $z(t) = (1/2 + a + it)/2$. The root-number phase is fixed by

$$\tau(\chi) = \sum_{r=1}^q \chi(r) e^{2\pi i r/q}, \quad \varepsilon_\chi = \frac{\tau(\chi)}{i^a \sqrt{q}}, \quad \alpha_\chi = -\frac{1}{2} \arg(\varepsilon_\chi).$$

For the principal zeta/L1 channel, $\alpha_\chi = 0$ and the two-channel eta readout uses the continuous phase

$$\phi_\eta(t) = \arg\left(1 - 2^{1-(1/2+it)}\right),$$

so that

$$\Theta_{L1}(t) = \vartheta_\zeta(t) + t \log C - \phi_\eta(t).$$

For non-principal χ ,

$$\Theta_\chi(t) = \vartheta_\chi(t) + t \log C + \alpha_\chi.$$

4 Harmonic Cost and the Quantum Ladder

The full source geometry is double ended. The midpoint is the origin of the fiber, and the reflected side is an anti-helix carrying the opposite orientation. The proof uses the positive oriented side of the helix for simplicity; the reflected anti-helix has the same cost law with the sign reversed.

The crossing levels in the projected unit-circle readout are therefore

$$A_k = \left(k + \frac{1}{2}\right) \pi, \quad k \in \mathbb{Z}.$$

Equivalently, for $m \geq 0$ the two oriented ladders are

$$A_m^+ = \left(m + \frac{1}{2}\right) \pi, \quad A_m^- = -\left(m + \frac{1}{2}\right) \pi.$$

On the positive helix side used in the proof,

$$A_0 = \frac{\pi}{2}, \quad A_{k+1} = A_k + \pi.$$

Thus the first crossing is the only crossing on that side whose cost from the origin is a half quantum:

$$|A_0^+| = \frac{\pi}{2}.$$

Every additional crossing on that same side costs one full quantum:

$$A_{m+1}^+ - A_m^+ = \pi \quad (m \geq 0).$$

The reflected anti-helix obeys the same rule:

$$|A_0^-| = \frac{\pi}{2}, \quad |A_{m+1}^- - A_m^-| = \pi \quad (m \geq 0).$$

So the half-quantum occurs once at the first crossing on each oriented side of the double-ended helix; all subsequent crossings consume full π increments.

Conjecture 4.1 (Origin/Tate interface). *If this helix model has a Tate-thesis-style global/local interpretation, the natural place for that connection is the shared origin of the double-ended helix. The origin is where the two reflected orientations meet before the one-dimensional readout separates them into positive and negative crossing ladders.*

5 Hilbert–Polya Infrastructure: R1–R9

This section lists the infrastructure available before the projection-faithfulness hypothesis is supplied. It does not assert GRH.

R1. Hilbert space. The source space is

$$\ell^2(\mathbb{N}; \mathbb{C}).$$

R2. Unitary phasor flow. The FTA/Euler phasor flow is

$$U(t)(n) = n^{it}, \quad |U(t)(n)| = 1,$$

with

$$U(s+t)(n) = U(s)(n)U(t)(n).$$

R3. **Real generator and no-drift principle.** The generator is

$$H(n) = \log n \in \mathbb{R}, \quad U(t)(n) = e^{itH(n)}.$$

The source no-drift lemma is: if

$$e^{\operatorname{Re} \lambda \tau} c = c \quad \text{for all } \tau, \quad c \neq 0,$$

then

$$\operatorname{Re} \lambda = 0.$$

R4. **Helix-native trace/readout equals L .** The finite phasor chain is

$$S_M(s, \chi) = \sum_{n \leq M} \chi(n) n^{-\operatorname{Re} s} U(\operatorname{Im} s)(n)^{-1}.$$

For $0 < \operatorname{Re} s$ and $\chi \neq 1$,

$$S_M(s, \chi) \rightarrow L(s, \chi)$$

with explicit rate:

$$\|S_M(s, \chi) - L(s, \chi)\| \leq C_s M^{-\operatorname{Re} s}.$$

R5. **Zeros induce fiber capture events.** At a zero $L(s, \chi) = 0$ with $0 < \operatorname{Re} s$,

$$\|S_M(s, \chi)\| \leq C_s M^{-\operatorname{Re} s}.$$

At every $\rho \in \operatorname{NTZ}(\chi)$,

$$-\frac{L'}{L}(s, \chi)$$

has no finite limit at ρ in the punctured neighborhood.

R6. **Spectral realization by crossing confluence and channel cancellation.** For any continuous strictly monotone accumulation E , if

$$E(a) \leq c \leq E(b), \quad a \leq b,$$

then there is a unique t such that

$$a \leq t, \quad E(t) = c.$$

At threshold $E(t) = n\pi$,

$$\left\lfloor \frac{E(t)}{\pi} \right\rfloor = n.$$

At a crossing threshold the source event is a confluence: the channel phase cancels, the amplitude reaches its crest, the sign changes, and the next harmonic begins on the spectral readout. The channel cancellation law is:

$$\sum_{m \in F} s_m = 0 \iff P_F(s) = M_F(s),$$

where

$$P_F(s) = \sum_{m \in F} \max(s_m, 0), \quad M_F(s) = \sum_{m \in F} \max(-s_m, 0).$$

At cancellation,

$$\sum_{m \in F} |s_m| = 2P_F(s).$$

R7. Exhaustion by ladder induction and rigidity. If a property holds at A_0 and is preserved by $A_k \mapsto A_{k+1}$ and $A_k \mapsto A_{k-1}$, then it holds at every A_k . Moreover, any crossing sequence c_n satisfying

$$E(c_n) = n\pi$$

is forced to be the purchase-height sequence:

$$c_n = \text{purchaseHeight}(n).$$

R8. Coordinate faithfulness and Pythagorean readout. The rechart

$$\text{arcChart}(x) = \frac{\pi}{3}x$$

has inverse

$$\text{arcChartInv}(u) = \frac{3}{\pi}u.$$

The critical midpoint is

$$\text{arcChart}(x) = \frac{\pi}{6} \iff x = \frac{1}{2}.$$

The midpoint conclusion routes through the unitary/Pythagorean readout. In natural coordinates the readout satisfies

$$(\text{readout}_{\sigma,\gamma})_{\text{Re}}^2 + (\text{readout}_{\sigma,\gamma})_{\text{Im}}^2 = 1 \iff \sigma = \frac{1}{2}.$$

Equivalently, for the spectral value $w(\rho) = 1 - 1/\rho$,

$$\|w(\rho)\|^2 = 1 \iff \text{Re } \rho = \frac{1}{2}.$$

The source conservation/no-drift input supplies the unitary source side of this readout.

R9. Genuine zero-set object. The object is Mathlib's L -function zero set:

$$\rho \in \text{NTZ}(\chi) \iff 0 < \text{Re } \rho < 1, \quad L(\rho, \chi) = 0.$$

6 Projection Faithfulness

Definition 6.1 (Source crossing confluence). *For a channel χ and point $\rho = \sigma + i\gamma$, a source crossing confluence at height γ is a three-dimensional fiber event with the following simultaneous features:*

- (C1) *the channel phases cancel, so the fiber vanishes in the channel sum;*
- (C2) *the accumulated amplitude reaches a crest at one of the ladder levels $A_m^\pm$;*
- (C3) *the crossing consumes the threshold energy required to start the next harmonic on the spectral readout;*
- (C4) *the real standing readout changes sign at the crossing;*
- (C5) *the source midpoint coordinate is $x = \pi/6$ in the $\pi/3$ chart, while the height γ is retained as the vertical coordinate.*

Under projection from the three-dimensional fiber to the unit-circle readout, radial and absolute phase information are collapsed. The retained datum is the crossing height γ , together with the unit-circle event. The subsequent one-dimensional readout rescales the midpoint coordinate by

$$x \mapsto \frac{3}{\pi}x,$$

so that the source midpoint $x = \pi/6$ is read as $1/2$.

Definition 6.2 (Source fiber event). *For a character χ and point ρ , a source fiber event is a source crossing confluence for χ at height $\text{Im } \rho$ whose source midpoint coordinate is $x = \pi/6$ in the $\pi/3$ chart. After the one-dimensional coordinate rescale $x \mapsto 3x/\pi$, its normalized midpoint readout is*

$$\text{Re } \rho = \frac{1}{2}.$$

Definition 6.3 (Fiber capture event). *For a Dirichlet channel χ ,*

$$\text{FiberCaptureEvent}_\chi(\rho) \quad :\Longleftrightarrow \quad \rho \in \text{NTZ}(\chi).$$

Hypothesis 6.4 (Faithful dimensional projection for one channel). *There is a predicate $\text{Proj}_\chi(\rho)$ of projected crossing events such that:*

(F1) *Every fiber capture event is faithfully projected as a crossing event:*

$$\rho \in \text{NTZ}(\chi) \implies \text{Proj}_\chi(\rho).$$

(F2) *Lower-dimensional projection introduces no false crossing events:*

$$\text{Proj}_\chi(\rho) \implies \rho \text{ is represented by a source fiber event.}$$

Thus the conditional input is only faithfulness of dimensional collapse at crossings: the projection does not miss fiber crossings carrying zero marks, and it does not create lower-dimensional events absent from the source. The midpoint implication itself is supplied by the unitary/Pythagorean readout described above.

7 Conditional Closure

Theorem 7.1 (Per-character conditional closure). *Assume $\text{FaithfulDimensionalProjection}$ for χ . Then*

$$\text{GRH}(\chi).$$

Proof. Let $\rho \in \text{NTZ}(\chi)$. By (F1), ρ is a projected fiber event. By (F2), that projected event is represented by a source fiber event. Source fiber events are registered at $x = \pi/6$ in the $\pi/3$ chart. The rechart $x \mapsto 3x/\pi$ and the Pythagorean unit readout give

$$\text{Re } \rho = \frac{1}{2}.$$

Since ρ was arbitrary, $\text{GRH}(\chi)$ follows. □

Theorem 7.2 (Complete conditional closure). *Assume faithful dimensional projection for every Dirichlet channel. Then*

$$\text{GRH}_{\text{complete}}.$$

Proof. Apply the per-character conditional closure to each modulus and character. $\square$

Theorem 7.3 (Zeta/L1 conditional closure). *Assume the corresponding eta-regularized zeta/L1 projection faithfulness. Then Mathlib's `RiemannHypothesis` holds.*

Proof. For $\rho \in \text{NTZ}_\zeta$, the zeta projection-faithfulness hypothesis gives a source fiber event. It is registered at $x = \pi/6$ in the $\pi/3$ chart, and the normalized Pythagorean readout is

$$\text{Re } \rho = \frac{1}{2}.$$

The existing zeta bridge converts

$$\forall \rho \in \text{NTZ}_\zeta, \quad \text{Re } \rho = \frac{1}{2}$$

to Mathlib's `RiemannHypothesis`. $\square$

8 Mathematical Reproduction of the Computation

The computation evaluates the source fiber equations above. It is not a proof step. The crossing ordinates are defined by the complete geometric fiber and its standing readout. Finite summation, residue decomposition, Hurwitz notation, and Euler–Maclaurin summation appear only where the derived fiber is represented for one-dimensional evaluation; they are not sources of the fiber.

8.1 Placed Geometric Fiber

The geometry is

$$A = \pi/6, \quad ds = \pi/3, \quad C = 2A \, ds = (\pi/3)^2.$$

For $s = \frac{1}{2} + it$, the placed finite head is

$$H_M(t) = \sum_{n=1}^M \chi(n) R_n^{-1} e^{-it \text{Log}(R_n^2)}, \quad R_n^2 = Cn.$$

Thus

$$H_M(t) = \sum_{n=1}^M \chi(n) (Cn)^{-s}.$$

The tail is not added from a separate external object. It is the remaining geometric fiber beyond the cutoff:

$$G_M(t) = \sum_{n>M} \chi(n) R_n^{-1} e^{-it \text{Log}(R_n^2)} = \sum_{n>M} \chi(n) (Cn)^{-s}.$$

The complete placed fiber is

$$F_\chi(t) = H_M(t) + G_M(t),$$

independently of the cutoff M when the tail is included.

8.2 Residue Coordinates for the Geometric Tail

The geometric tail may be written in conductor residue coordinates. Let $n_{0,r}$ be the first integer $> M$ in residue class $r \bmod q$. Then

$$G_M(t) = \sum_r \chi(r) C^{-s} q^{-s} \sum_{m \geq 0} \left(\frac{n_{0,r}}{q} + m \right)^{-s}.$$

This identity is just the residue-class reindexing of the geometric terms $R_n^2 = Cn$. Hurwitz notation abbreviates the inner residue tail; it does not define the tail or the crossing condition. It is a coincidental representation of the derived fiber.

For finite one-dimensional evaluation, replace the inner infinite residue tail by

$$\mathcal{T}_{h,J}(s, x) \approx \sum_{m \geq 0} (x + m)^{-s}.$$

For integers $h, J \geq 1$, set $X = x + h$ and write

$$(s)^{\overline{m}} = s(s+1) \cdots (s+m-1), \quad (s)^{\overline{0}} = 1.$$

The Euler–Maclaurin summation evaluator used for that geometric residue tail is

$$\begin{aligned} \mathcal{T}_{h,J}(s, x) &= \sum_{m=0}^{h-1} (x + m)^{-s} + \frac{X^{1-s}}{s-1} + \frac{1}{2} X^{-s} \\ &\quad + \sum_{j=1}^J \frac{B_{2j}}{(2j)!} (s)^{\overline{2j-1}} X^{-s-2j+1}, \end{aligned}$$

where B_{2j} are Bernoulli numbers. The associated remainder is the standard Euler–Maclaurin remainder after the B_{2J} term. The numerical tables in the repository use $h = 8$ and finite J chosen large enough for the working precision.

With this evaluator, the finite computed fiber is

$$F_{M,\chi}(t) = \sum_{n=1}^M \chi(n) (Cn)^{-s} + \sum_r \chi(r) C^{-s} q^{-s} \mathcal{T}_{h,J} \left(s, \frac{n_{0,r}}{q} \right).$$

For the L1 channel, the same geometric tail is grouped with conductor 2 and weights

$$\chi_\eta(1) = 1, \quad \chi_\eta(0) = -1,$$

so that

$$F_{M,L1}(t) = C^{-s} 2^{-s} \left[\mathcal{T}_{h,J} \left(s, \frac{n_{0,1}}{2} \right) - \mathcal{T}_{h,J} \left(s, \frac{n_{0,0}}{2} \right) \right] + \sum_{n=1}^M (-1)^{n+1} (Cn)^{-s}.$$

This is the finite evaluation of the eta-regularized zeta fiber in placed geometric coordinates.

8.3 Principal One-Dimensional Tail Error Control

For the projected principal/L1 tail, the repository also formalizes the classical Euler–Maclaurin summation identity as an error-control theorem. Let

$$S_N(s) = \sum_{n=1}^N n^{-s}, \quad \{u\} = u - \lfloor u \rfloor,$$

and define

$$I(s, N) = \int_N^\infty \{u\} u^{-s-1} du, \quad R(s, N) = sI(s, N).$$

For $\operatorname{Re} s > 0$ and $s \neq 1$, the continued one-dimensional response is

$$c(s) = \frac{s}{s-1} - s \int_1^\infty \{u\} u^{-s-1} du,$$

and the Euler–Maclaurin identity is

$$c(s) = \zeta(s),$$

with finite-readout error

$$S_N(s) - \zeta(s) - \frac{N^{1-s}}{1-s} = R(s, N)$$

and bound

$$\|R(s, N)\| \leq \frac{\|s\|}{\operatorname{Re} s} N^{-\operatorname{Re} s} \quad (N \geq 2).$$

This theorem controls the finite projected evaluation of the principal geometric tail; it is a representation of the derived fiber, not a source for the tail.

The full finite readout is

$$Z_\chi(t) = \operatorname{Re}(e^{i\Theta_\chi(t)} F_{M,\chi}(t)).$$

Crossings are isolated by sign changes

$$Z_\chi(a)Z_\chi(b) < 0$$

followed by bisection. If $\gamma \in (a, b)$ is a simple crossing,

$$Z_\chi(\gamma) = 0, \quad Z'_\chi(\gamma) \neq 0,$$

then there is an interval around γ on which the sign changes once, and bisection on any bracket inside that interval converges to γ .

For local reproduction from an intermediate height t_0 , choose $\delta > 0$ and evaluate Z_χ on $[t_0 - \delta, t_0 + \delta]$. If a sign change occurs in $[a, b] \subset [t_0 - \delta, t_0 + \delta]$, bisection on Z_χ rederives the crossing in that local interval; no earlier crossing is used.

9 Formal and Computational Support

The proof above is the mathematical statement. The repository support has two roles: the formal files check the stated implications in Lean, and the numerical files evaluate the displayed fiber equations.

9.1 Formalization Map

Dirichlet zero set	GRHSpectral.NontrivialZeros
per-character GRH predicate	GRHSpectral.GRH
all-character GRH predicate	GRHSpectral.GRHComplete
zeta zero set	ZD.NontrivialZeros
Euler–Maclaurin continuation	GRH.EulerMaclaurinDirichlet.c_fun
Euler–Maclaurin error term	GRH.EulerMaclaurinDirichlet.R
Euler–Maclaurin identity	GRH.EulerMaclaurinDirichlet.difference_eq_R
Euler–Maclaurin bound	GRH.EulerMaclaurinDirichlet. euler_maclaurin_dirichlet_explicit
zeta one-dimensional completion	SourceContinuation.zetaCompletion
residue fiber decomposition	DirichletFibre.LFunction_eq_fibre_sum
fiber cancellation	DirichletFibre.zero_iff_fibre_cancellation
source no-drift	HelixSource.source_noDrift
Möbius unit readout	moebius_unit_iff
Pythagorean readout	readout_pythagoras_iff_online
spectral unit midpoint	SpectralSide.w_unit_iff_half
$\pi/3$ gauge	HelixGauge.piThirdGauge_eq
nonprincipal closed form	HelixGauge.piThirdHelixSource_eq_gauge_mul_L
L1 closed form	HelixGauge.piThirdZetaFiber
L1 zero equivalence	HelixGauge.piThirdZetaFiber_zero_iff
R1–R9 infrastructure	hilbertPolyaChainR1ToR9
projection faithfulness	FaithfulDimensionalProjection
per-character closure	GRH_of_HP_and_Faithfulness
all-character closure	GRHComplete_of_HP_and_Faithfulness
zeta/L1 closure	RH_of_HP_and_Faithfulness

9.2 Lean Verification Protocol

File-level verification:

```
lake env lean RequestProject/HilbertPolyaChain.lean
```

Closed-form gauge verification:

```
lake env lean RequestProject/HelixGaugeIdentity.lean
```

Escape-hatch audit:

```
rg -n "axiom |sorry|admit|@[implemented_by\]" \
RequestProject/HilbertPolyaChain.lean
```

Axiom footprint targets:

```
#print axioms hilbertPolyaChainR1ToR9,
#print axioms GRH_of_HP_and_Faithfulness,
#print axioms GRHComplete_of_HP_and_Faithfulness,
#print axioms RH_of_HP_and_Faithfulness,
#print axioms HelixGauge.piThirdGauge_eq,
#print axioms HelixGauge.piThirdZetaFiber_zero_iff.
```

Expected footprint:

```
[propept, Classical.choice, Quot.sound].
```

10 Current Logical Status

HP R1–R9 infrastructure	Lean theorem	<code>hilbertPolyaChainR1ToR9,</code>
$\pi/3$ closed form	Lean theorems	<code>HelixGauge.piThird*,</code>
Projection faithfulness	hypothesis	<code>FaithfulDimensionalProjection,</code>
Per-character closure	conditional theorem	<code>GRH_of_HP_and_Faithfulness,</code>
All-character closure	conditional theorem	<code>GRHComplete_of_HP_and_Faithfulness,</code>
Zeta/L1 closure	conditional theorem	<code>RH_of_HP_and_Faithfulness.</code>

The work required to remove the condition is precisely a proof of projection faithfulness: fiber capture events of the FTA-generated source fiber must survive the dimensional projection to the lower-dimensional readout, and the projection must not introduce events absent from the source.